

KRITYA RAJANSH

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[Kritya Rajansh](#)



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EDUCATION

Indian Institute of Information Technology, Kalyani
B.Tech, Computer Science & Engineering

2023 – 2027

9.69 CGPA

Bishop Scott Boys' School, Patna, Bihar
Intermediate (91%) & Matriculation (98.6%)

SKILLS

Languages	Python, JavaScript , C, SQL
Web	React.js, Node.js, Express.js, Flask, FastAPI, REST API design, JWT
Databases	MongoDB (Mongoose, \$text indexing), MySQL
ML / AI	PyTorch, Scikit-learn, NumPy, Pandas, TF-IDF, RAG pipelines
Concepts	DSA, OOP, DBMS, asyncio, Git

EXPERIENCE

Research Intern – DRDO Sponsored Project
IIIT Kalyani

March 2026 – May 2026

Kalyani, West Bengal

- Designed and implemented an N-node Kademlia DHT simulation from scratch, modelling distributed key-value storage across a variable-size peer network with TLS 1.3 mutual authentication at the transport layer.
- Engineered realistic $O(\log N)$ lookup behaviour through staggered bootstrapping and sparse routing tables (k-bucket size $K=6$), and validated availability degradation under targeted churn against data-holding nodes.
- Conducted experiments measuring lookup hop scaling across $N=10$ to 200 nodes and key availability under churn fractions up to 70%, producing results consistent with Kademlia's theoretical guarantees.
- [Internship Certificate](#)

PROJECTS

Doc-Doctor: Enterprise Search Assistant

- Built a full-stack MERN application with a Python/Flask microservice for enterprise document search, implementing TF-IDF ranking and RAG-based answer synthesis via the Anthropic API.
- Tech:* React.js, Node.js, Express.js, MongoDB, Python, Flask, JWT

SAR Image Processing & Enhancement

- Benchmarked five SAR reconstruction algorithms (MF, RMA, FA-PnP-ADMM, PnP-CNN-ADMM, PnP-AE-ADMM) with quantitative fidelity comparisons; engineering an Autoencoder-based denoising stage to suppress speckle noise beyond classical baselines.
- Tech:* Python, PyTorch, Scikit-learn, NumPy, Matplotlib, Jupyter Notebook

P2P Full Stack Simulation: 2-Node Deep Dive

- Extended N-node DHT to a 2-node full-stack simulation exposing every protocol layer: ASK/FSK/PSK modulation with Gaussian noise, CRC-32 framing, stop-and-wait ARQ, and IPsec vs. mTLS overhead benchmarking across message sizes 32B–4KB.
- Tech:* Python, asyncio, ssl (TLS 1.3), cryptography, Jupyter Notebook